TEAM SEAT 01 OF 04 · INSTITUTIONAL QUANTITATIVE DESK OPERATING MANUAL

FINANCE, PORTFOLIO & RISK OFFICER HANDBOOK

Role Mandate: Capital Preservation, Position Sizing, Friction Auditing & P&L Governance System: trading-sandbox v2.4

YOUR MISSION INSIDE THE 4-PERSON QUANTITATIVE TEAM

As the **Finance & Risk Officer (Seat 01)**, you are the guardian of the firm's balance sheet ($400,000.00 Alpaca Paper Sandbox and segregated live capital). You do not write C++/Python infrastructure or train machine learning trees—instead, you set the hard risk limits in `.env`, audit strategy net returns after transaction friction, approve or reject models proposed by the **Quant Researcher (Seat 02)**, verify automated circuit breakers with the **Software Engineer (Seat 03)**, and oversee daily ledger reconciliations with the **Operations Specialist (Seat 04)**.

1.1 CAPITAL SEGREGATION & THE 4-TIER RISK CIRCUIT BREAKER

In retail trading, 80%–95% of participants lose money due to emotional over-leveraging and unquantified transaction costs. Our desk enforces strict capital segregation (long-term investment wealth is never mingled with active trading equity) and requires a mandatory 1-to-2 month live simulation inside our **$400,000.00 Alpaca Paper Sandbox** before live deployment. Every trade signal must pass through `RiskEngine` (`risk_manager.py`), which enforces your hard-coded mathematical limits:

Institutional Share Sizing Law: $Shares = min(\lfloor (Equity \times 0.01) / |P_{entry} - P_{stop}| \rfloor , \lfloor (Equity \times 0.05) / P_{entry} \rfloor)$ | **Half-Kelly Leverage:** $f^*_{half} = 0.5 \times (p - q / b)$

Risk Parameter	System Variable / Function	Mandated Limit	Financial Purpose & Dollar Cap on $400,000 Equity
Per-Trade Stop-Loss Cap	`MAX_RISK_PER_TRADE=0.01`	**1.00% Max Loss**	Caps maximum realized loss at **$4,000.00** if the technical stop-loss is hit. At $150 entry and $145 stop ($5 risk/share), risk math permits 800 shares.
Notional Concentration Cap	`MAX_POSITION_SIZE=0.05`	**5.00% Exposure**	Caps single-ticker market exposure at **$20,000.00**. In the $150 entry example, 800 shares ($120,000 notional) is automatically throttled down to **133 shares** ($19,950).
Daily Drawdown Kill Switch	`MAX_DAILY_DRAWDOWN=0.03`	**3.00% Intraday Halt**	If live equity drops **$12,000.00** below `start_of_day_equity`, `check_daily_circuit_breaker()` locks order routing for the day and alerts Discord.
Single-Strategy Half-Kelly	`half_kelly_fraction(p, b)`	$0.5 \times (m / s^2)$	Protects compounded growth rate $g = r + S^2/2$ against fat-tailed kurtosis (Black Swan crashes). Full Kelly is strictly prohibited on our desk.
Multi-Strategy Covariance	Portfolio Allocation Matrix	$F^* = 0.5 \times C^{-1}M$	Allocates capital across uncorrelated strategies using inverse return covariance C^{-1} and mean excess return vector M, scaling down immediately after losses.
Microstructure & Liquidity	Execution Gatekeeper	Price > $5.00 Size < 1% ADV	Bans illiquid penny stocks (<$5.00) with punitive percentage spreads and restricts order size below 1% of Average Daily Volume (ADV) to prevent market impact.

1.2 STRUCTURAL EDGE: WHY OUR PRIVATE DESK BEATS GIANT HEDGE FUNDS

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1.3 EMPIRICAL P&L AUDIT: THE SPY REGIME MISMATCH & FRICTION PROOF

Before approving any strategy submitted by the Quant Researcher (Seat 02), you must audit its net performance after deducting **8 basis points (0.08% / `spread_slippage_bps = 0.0008`)** per turnover event. High-turnover strategies that claim +5 bps gross per trade while paying −8 bps in bid-ask spread and slippage lose −3 bps on every execution. Review the live audit below from `quant_engine.py` on **SPY**:

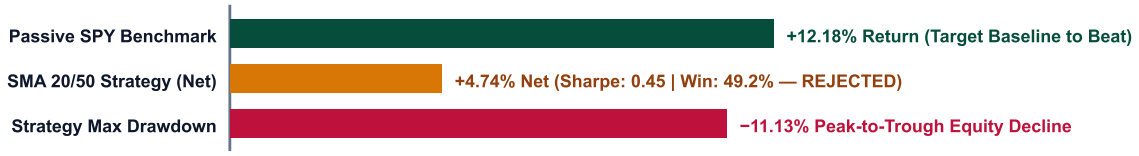


Figure 1.1: Why Finance Rejected Unfiltered SMA 20/50 on SPY — Deploying momentum inside a Mean-Reverting Regime (*H* = *0.4647*) caused whipsaw losses and 8 bps turnover drag, trailing passive SPY by −7.44%.

1.4 DAILY FINANCIAL WORKFLOW: WEB UI TAB 3 & DISCORD #RISK

Your daily operational routine uses two specialized interfaces that require zero code editing:

- **Discord `#risk` Pre-Trade Sizing Check:** Type `/position_size entry:150 stop:145` to invoke `cogs/modelling.py`. Verify that the blue Risk Management Embed matches your 1% risk ($4,000) and 5% notional ($20,000) caps. Note: In mean-reverting regimes, ensure technical stops are set wide enough to avoid locking in temporary liquidity shocks right before price snaps back to the mean.
- **Web Dashboard (`index.html` Tab 3 · Execution Spreadsheet):** Audit all 15 institutional ledger columns (Columns A–O: Order ID, Timestamp, Ticker, Strategy, Regime *H* & ADF *p*, Side, Qty, Entry, Exit, Z-Score, ML Confidence, Friction bps, Net P&L, and Quant Audit Notes). Inspect the live formula bar (`=SUM(Net_PnL)`, `=WIN_RATE()`, `=AVERAGE(Slippage)`) and click ⬇ **Export .CSV (Excel)** for weekly accounting reconciliation.

1.5 CROSS-TEAM HANDOFF PROTOCOL: HOW SEAT 01 WORKS WITH SEATS 02–04

Team Counterpart	What You Receive From Them	Your Sign-Off Gate & Deliverable Back to the Team
Seat 02: Quant / Math Researcher	Out-of-sample backtest matrices (`#backtests`) and Purged CV Meta-Labeling accuracy reports.	**Capital Sign-Off Gate:** Approve for paper/live allocation *only* if Annualized Sharpe > 2.0 (min > 1.5), Max Drawdown < 10%–15%, Free Parameters ≤ 5, and Net Return beats SPY after 8 bps friction.
Seat 03: Software Engineer	Automated `pytest` pass logs (`tests/test_risk_manager.py`) and cloud `ExecutionOMS` fill webhooks.	**Risk Parameter Specs:** Provide updated `MAX_RISK_PER_TRADE`, `MAX_POSITION_SIZE`, and `MAX_DAILY_DRAWDOWN` constants for `.env`. Never allow merges if risk unit tests fail.
Seat 04: General Operations	Daily anomaly flags from Web UI Tab 1 logs and Discord `#system-status` monitoring.	**Emergency Authorization:** Authorize Seat 04 to trigger `!cancelall` if broker fills diverge from backtested slippage expectations or during macro black-swan events.

SHARPE RATIO FINANCING NUANCE FOR PORTFOLIO ACCOUNTING

When auditing strategy Sharpe ratios ($SR = \sqrt{N_T} \times R / s$), subtract the risk-free rate r_F **only** if the strategy incurs overnight financing costs (such as long-only swing positions). For dollar-neutral long-short pairs (e.g., `GLD/GDX` where short proceeds fund the long leg) or intraday-only strategies, the portfolio is self-financing—do not subtract r_F.